

**Consumer Heterogeneity in Online Reviews:
How Average Consumption Profiles Influence Evaluative
Disagreement**

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Abstract

Online ratings are often assessed as objective indicators of quality, yet they are frequently influenced by unobserved consumer heterogeneity. This paper examines how consumer expectations formed through past experiences influence these evaluations. Using a large-scale dataset on TripAdvisor restaurant reviews, we construct reviewer pairs who evaluated the same establishment but differ in their average consumption profiles (average price level of restaurants reviewed). Reviewers accustomed to more expensive restaurants systematically assign lower ratings to the same establishments. The research also demonstrates that this effect remains significant after controlling for public online aggregate ratings which act as reputation signals, and the clientele heterogeneity which the restaurant attracts, as they also play critical roles in shaping evaluative disagreements. The project advances theoretical discussions on what ratings measure and provides actionable insights into whether ratings reflect true quality versus consumer-product fit.

Keywords:

Online reviews, Evaluative disagreement, Consumer heterogeneity

INTRODUCTION

Online review platforms have become a central part of consumer decision-making. Online reviews serve as a powerful form of electronic word-of-mouth (eWOM), where reviews enhance consumer trust, reduce perceived risk, and increase purchase likelihood when perceived as credible and detailed (Choudhary, 2025).

Consumers increasingly rely on ratings and reviews when making purchase decisions, making online evaluations an important determinant of market demand. Empirical studies find that favourable reviews increase consumer demand, sales, and firm revenues across a variety of contexts (Zhu & Zhang, 2010; Luca 2011). For example, a one-point increase in average ratings is associated with a 4% increase in video game sales (Zhu & Zhang, 2006), and a half-star increase in Yelp ratings causes restaurants to sell out 49% more frequently (Anderson & Magruder, 2012). Online review platforms accelerate consumer learning about service or product quality and also shape industry revenues (Fang, 2022), which makes understanding how online evaluations are formed an important question for both researchers and managers.

Relatively little research has examined why consumers systematically disagree when evaluating the same experience. Some disagreement may arise because consumers encounter objectively different service conditions. In the context of restaurants, diners may visit at different times of day, interact with different staff members, be seated in different locations, order different menu items, or visit when service quality and restaurant crowding differ. These factors can generate genuinely different consumption experiences and, consequently, different evaluations. Yet disagreement may also persist even when consumers encounter broadly similar conditions. For example, consumers subjected to higher prices often anticipate a greater level of quality (Jeziorski & Michelidaki, 2023). As a result, variation in online ratings may reflect not only differences in the quality delivered but also differences in how consumers assess that quality.

Previous studies have not directly compared evaluations from two different consumers observing the same product or service. This study addresses this gap using a large-scale data set of restaurant reviews. We construct pairs of reviewers who evaluated the same restaurant and examine whether differences in reviewers’ prior consumption profiles predict systematic differences

in ratings. Reviewer consumption profiles are measured using the average price level of restaurants each reviewer visits, allowing us to capture variation in prior experiences and expectations. By comparing reviewers evaluating the same restaurant, the design isolates heterogeneity in consumer evaluations while holding the restaurant constant. Beyond reviewer average price differences, this study investigates whether rating disagreement varies according to changes in a restaurant’s online reputation signal, captured by the estimated aggregate rating at the time of review, and the heterogeneity of the restaurant’s customer base. The same experience may generate very different evaluations across consumers, raising questions about whether average ratings accurately reflect underlying quality.

The consequences of evaluative disagreements may extend beyond individual evaluations. Because many online platforms rank and display products according to aggregate ratings and popularity metrics, even relatively small changes in average ratings can influence a product’s visibility and consumer demand. Airbnb for example, communicate that even a star rating of 4.3/5 may lead to a listing suspension (Jeziorski & Michelidaki, 2023). Consequently, maintaining favourable online ratings has become a strategic priority for firms. Yet despite the growing importance of these evaluative systems, relatively little research has examined whether consumers truly use common standards when generating the ratings on which these systems depend.

CONTRIBUTIONS

This study contributes to the online reviews and consumer behaviour literature in several ways. First, this study provides an explanation for disagreement in online ratings by linking consumer evaluations to prior consumption experiences and reference standards. While existing research has documented the consequences of online reviews for demand and firm performance (Zhu & Zhang., 2010; Zhu & Zhang 2006; Luca, 2011) considerably less attention has been devoted to understanding why consumers evaluating the same experience often arrive at different conclusions. This study argues that reviewer disagreement emerges not only from varying product or service experiences, but also because consumers evaluate experiences relative to different expectations and standards formed through their prior consumption histories.

Second, this paper introduces a novel pairwise reviewer design to the online reviews literature.

Rather than focusing on aggregate ratings alone, we construct pairs of reviewers who evaluated the same restaurant and compare their ratings based on differences in their average consumption patterns.

Third, the study extends expectation-disconfirmation (Oliver, 1980) and reference point theories (Mazumdar, Raj, & Sinha, 2005) into the context of digital review platforms. We show that consumers accustomed to different types of restaurants may interpret and evaluate identical experiences differently because they compare experiences to different internal standards. In doing so, the paper demonstrates that online ratings may reflect heterogeneous expectations and subjective evaluative frameworks rather than purely objective assessments of quality.

Finally, the paper contributes to the broader literature on online review platforms and digital reputation systems by showing that aggregate ratings may conceal substantial disagreement across consumer types. Our findings suggest that disagreement and variance in online reviews contain economically meaningful information beyond average ratings alone, with implications to managerial strategies like customer targeting. For example, restaurants attracting highly heterogeneous customers may experience greater variation in ratings because consumers evaluate the same experience using different reference standards and expectations. Managers may therefore benefit from aligning marketing communications, pricing, and positioning strategies with their intended clienteles in order to reduce large expectation differences and generate more consistent evaluations. More broadly, the findings suggest that firms and review platforms should pay attention not only to average ratings, but also to patterns of disagreement across customer groups, as these may provide valuable information about customer fit and the consistency of their customer’s experiences and furthermore, public evaluations.

LITERATURE REVIEW

Expectations, Reference Points, and Consumer Evaluation

Consumer evaluations are rarely based on a universal or objective standard of quality (Pasaco-González et al., 2023). Consumers rely on category schemas and prior knowledge when forming expectations, and that evaluations depend not only on product attributes themselves

but also on the extent to which the experience confirms or disconfirms those expectations (Stayman, Alden, & Smith, 1992). Tourism and consumer research suggest that experiences are inherently subjective, with individuals assigning different meanings to the same event based on their personal characteristics and accumulated experiences (Pasaco-González et al., 2023), while consumers unconsciously adjust evaluative standards based on brand expectations rather than applying consistent criteria (Dimofte & Johansson, 2009). Expectation-disconfirmation theory proposes that satisfaction depends on the gap between expected and actual performance, such that consumers report higher satisfaction when experiences exceed expectations and lower satisfaction when experiences fall short of them (Oliver, 1980). Consequently, two consumers experiencing the same product or service may arrive at different evaluations if they begin with different expectations. Relatedly, reference point theory argues that consumers evaluate products relative to internal standards developed through previous experiences (Mazumdar, Raj, & Sinha, 2005). These reference standards influence how consumers interpret quality and value. Individuals with extensive experience in premium consumption environments may therefore assess a restaurant differently than individuals whose prior experiences are concentrated in lower-priced establishments. Together, these perspectives suggest that consumers may apply different standards when judging the same experience.

Limitations of Aggregate Ratings

Aggregate ratings summarize the evaluations of many consumers into a single numerical score, and then are displayed on review platforms such as Tripadvisor, Yelp, Airbnb, and Amazon for public visibility. Despite their widespread use, aggregate ratings rest on a strong assumption: that evaluations produced by different consumers can be meaningfully combined into a single measure of quality. Aggregate ratings obscure the experiences, preferences, and evaluative standards of individual reviewers, and rather are regarded as an impression of the overall consumer opinion (Ziegele & Weber, 2015). By averaging reviews across individuals, rating systems implicitly assume that consumers assess products and services using comparable standards. Yet recent research suggests that aggregate ratings may not always provide an accurate representation of underlying quality. Average ratings correlate only weakly with objective measures of quality and can be influenced by factors such as price, brand reputation, reviewer selection, and differences in consumer

preferences (De Langhe, Fernbach, & Lichtenstein, 2016; Li & Hitt, 2008). Consequently, averaging consumer evaluations can obscure important differences across reviewers and introduce systematic biases into rating systems. Despite these limitations, consumers place substantial weight on average ratings when making quality judgments (De Langhe, Fernbach, & Lichtenstein, 2016), creating what the authors describe as an “illusion of validity.” Their findings suggest that consumers often interpret aggregate ratings as objective indicators of quality even when ratings may reflect other influences unrelated to actual performance. Dai et al. (2012) show that average online ratings can deviate substantially from optimal ratings because aggregation mechanisms fail to fully account for differences among reviewers. Similarly, Li and Hitt (2008) argue that online review systems may be affected by self-selection biases, as consumers with particularly strong experiences are more likely to post reviews. These dynamics can cause average ratings to evolve over time in ways that do not necessarily reflect objective changes in product quality. Online ratings are influenced by preference heterogeneity and social learning processes (Besbes and Scarsini, 2018), suggesting that average ratings may not accurately represent underlying quality when consumers evaluate experiences differently. However, while prior research identifies several limitations of rating aggregation, relatively little work has directly examined how differences in consumers themselves contribute to evaluative disagreement.

Consumer Heterogeneity and Subjective Evaluations

Consumer disagreement may arise because quality itself is multidimensional (Hjorth-Andersen, 1984). Marketing research distinguishes between vertical differentiation, where consumers largely agree on quality rankings, and horizontal differentiation, where evaluations depend on individual tastes and preferences (Gabszewicz & Thisse, 1986). Under vertical differentiation, consumers may disagree about how much they are willing to pay for quality, but they generally agree about which products are superior. Under horizontal differentiation, however, consumers can legitimately disagree because different product attributes appeal to different consumers. Experience goods are particularly susceptible to horizontal differentiation because consumers often place different importance on subjective attributes such as ambiance or aesthetics. Adhikari (2015) distinguishes between objective and subjective attributes of experience goods, arguing that consumers frequently evaluate experiential characteristics differently. For example, consumers interpret quality signals

such as local origin, organic certification, or even electric vehicles in distinct ways depending on their preferences and backgrounds (Di Vita et al., 2021; Jaiswal et al., 2022), further supporting the idea that one quality signal can mean two different things to different consumers. This heterogeneity extends to online review environments: consumers respond differently to negative reviews depending on identity relevance and social distance (Ordabayeva et al., 2022), and review effects vary across consumer segments and according to the attributes emphasized in reviews (Cho et al., 2024). While prior research establishes that consumers differ in their preferences, evaluations, and responsiveness to information, relatively little research has examined how these differences translate into systematic disagreement in online ratings. As a result, it remains unclear whether aggregate ratings obscure meaningful variation in how different consumers assess the same experience.

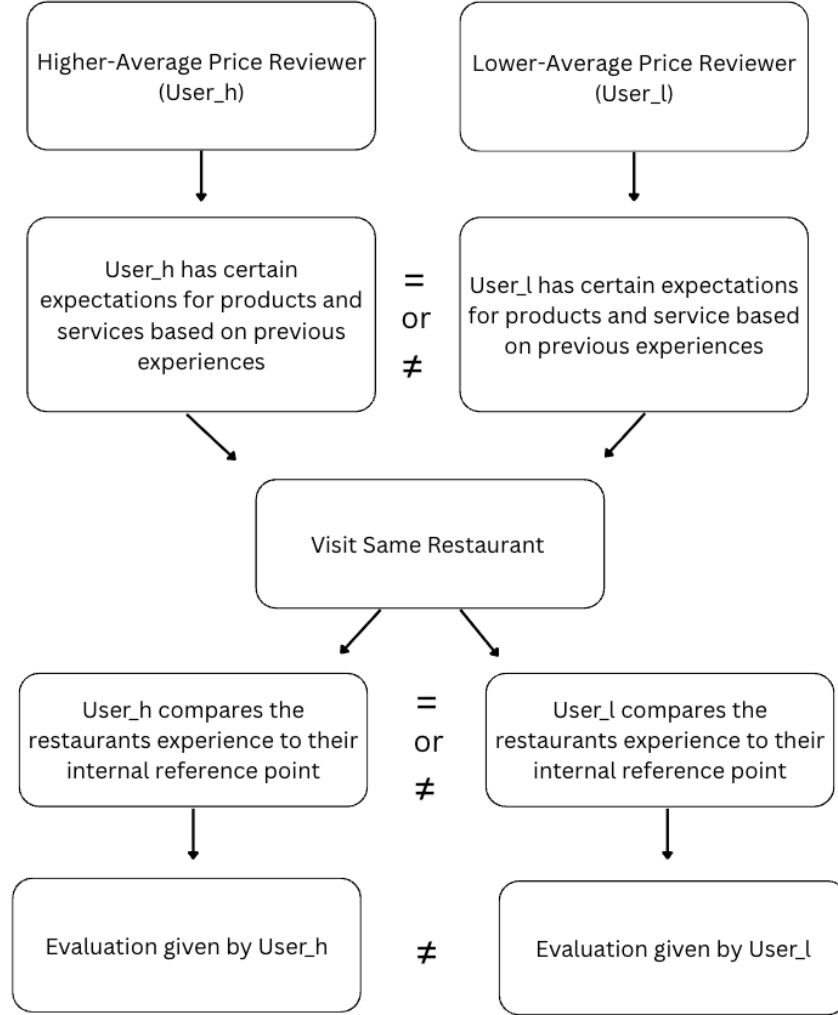
Addressing the Gap

We know relatively little about whether disagreement in online ratings reflects systematic differences in evaluative standards and prior consumption experiences rather than differences in objective quality alone. This study addresses this gap by constructing a large-scale reviewer-pair dataset that compares consumers who evaluated the same restaurant but differ in their prior consumption profiles. By focusing on evaluations of an identical restaurant, the approach allows differences in ratings to be linked more directly to differences in reviewers' prior experiences. In doing so, the study contributes to our understanding of how heterogeneous evaluative frameworks shape online ratings, or more so the disagreements between them.

CONCEPTUAL FRAMEWORK

Consumers may differ substantially in their average consumption profiles. Some consumers regularly visit higher-priced restaurants, while others primarily visit lower-priced ones. These differences in prior consumption experiences may expose consumers to different levels of service, food quality, atmosphere, and other restaurant attributes, thereby contributing to the formation of different expectations and evaluative reference points.

Figure 1: Conceptual Framework



According to expectation-disconfirmation theory, consumers evaluate experiences by comparing actual performance with their prior expectations (Oliver, 1980). Over time, as consumers gain experience through their consumption practices, they establish internal reference points against which they naturally compare new products or services to (Mazumdar, Raj, & Sinha, 2005).

Consumers may then enter the same restaurant with completely different expectations. These expectations are shaped by prior restaurant experiences and may also be influenced by other sources of information, such as word of mouth or through observing online ratings and reviews posted online beforehand. After visiting the restaurant, each consumer compares the experience with their own

internal reference point and evaluates it based on whether the experience fell short of, met, or exceeded what they had expected. Consequently, consumers evaluating the same restaurant may arrive at different evaluations, in part because they assess their experiences relative to different expectations and internal reference points, rather than solely differences in their restaurant experience.

Ratings therefore reflect not only the characteristics of the service encounter itself but also consumers' subjective evaluations shaped by their prior consumption profiles, reference points, and expectations. This perspective helps explain why evaluative disagreement can emerge among consumers evaluating the same restaurant and suggests that aggregate ratings may capture differences in evaluative standards rather than solely differences in underlying quality.

METHOD/DATA

Data Source

For this study we use a large-scale data set of online restaurant reviews collected from TripAdvisor. Reviewer and restaurant identifiers, review dates, numerical ratings on a 10–50 scale (with 50 representing the highest possible rating), and restaurant characteristics such as city, cuisine type, and price category are all provided. For the main analysis, the study focuses specifically on restaurants located in Barcelona, Spain. The sample is further restricted to reviews posted from January 1, 2013 through January 1, 2020.

Sample Construction

We construct the sample to include only active reviewers and sufficiently reviewed restaurants. We restrict the sample to reviewers with at least 10 reviews to reduce concerns that ratings reflect only special-occasion experiences rather than stable reviewing behaviour. Prior research suggests that consumers with extremely positive or extremely negative experiences are more likely to engage in word-of-mouth behaviour (Anderson, 1998), meaning infrequent reviewers may disproportionately reflect atypical experiences. The sample is also restricted to restaurants with at least 10 reviews to ensure sufficient review activity within each restaurant for meaningful reviewer

comparisons and more reliable measures of reviewer disagreement. Additionally, only the first restaurant review is kept per user if they have visited the establishment more than once.

The final dataset contains 124,175 reviews written by 7,067 unique users across 2,728 restaurants.

The central idea of this study is to compare how two different consumers evaluate the same establishment. We construct a reviewer-pair dataset by matching all unique users who evaluated the same restaurant. This reviewer-pair construction represents a key methodological contribution of the study, as it allows systematic disagreement between reviewers with different prior consumption trends. The final reviewer-pair dataset contains 6,503,593 reviewer pairs.

Metrics

For each restaurant j , pairs of reviewers h and l are constructed who both evaluated the restaurant. $User_h$ is defined as the reviewer whose average restaurant price level is higher, while $User_l$ is defined as the reviewer whose average restaurant price level is lower. Meaning $User_h$ typically visits higher-priced restaurants relative to $User_l$ who typically visits lower-priced restaurants, based on their reviewing history. Importantly, a reviewer who appears as $User_h$ in one pair may appear as $User_l$ in another pair depending on the relative average price levels of the two reviewers being compared.

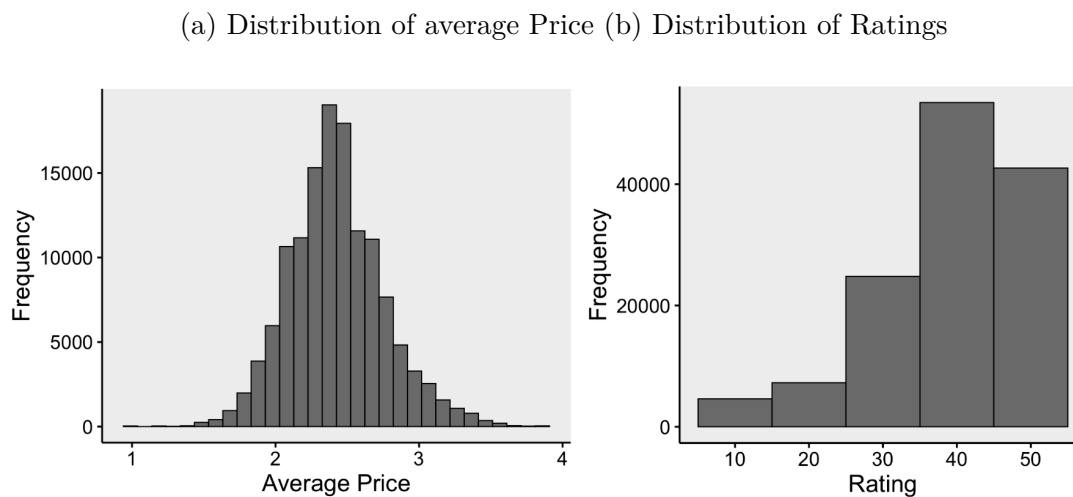


Figure 2: Distributions of reviewer average prices and ratings

Figure 2 presents the distributions of reviewer average prices and ratings. Panel (a) shows the distribution of all reviewers' average price levels, while Panel (b) shows the distribution of ratings.

To capture differences in reviewers' average price standards:

$$\Delta AveragePrice_{hl} = AveragePrice_h - AveragePrice_l$$

where $AveragePrice_h$ and $AveragePrice_l$ represent the average price levels associated with reviewers h and l previous restaurant visits, respectively.

The dependent variable is the difference in the ratings assigned by the two reviewers:

$$\Delta Rating_{hlj} = Rating_{hj} - Rating_{lj}$$

where $Rating_{hj}$ denotes the rating assigned by reviewer h to restaurant j , and $Rating_{lj}$ denotes the rating assigned by reviewer l to restaurant j .

In figure 3, positive values indicate cases in which higher-average price reviewers assigned higher ratings, while negative values indicate cases in which higher-average price reviewers assigned lower ratings. The distribution is centered slightly below zero, suggesting that reviewers who visit more expensive restaurants tend to evaluate the same establishments somewhat more negatively on average.

Figure 4 shows a clear negative relationship between $\Delta AveragePrice_{hl}$ and $\Delta Rating_{hl}$. As the difference in reviewers' average restaurant price levels increases, the higher-average price reviewer tends to assign increasingly lower ratings relative to the lower-average price reviewer evaluating the same restaurant.

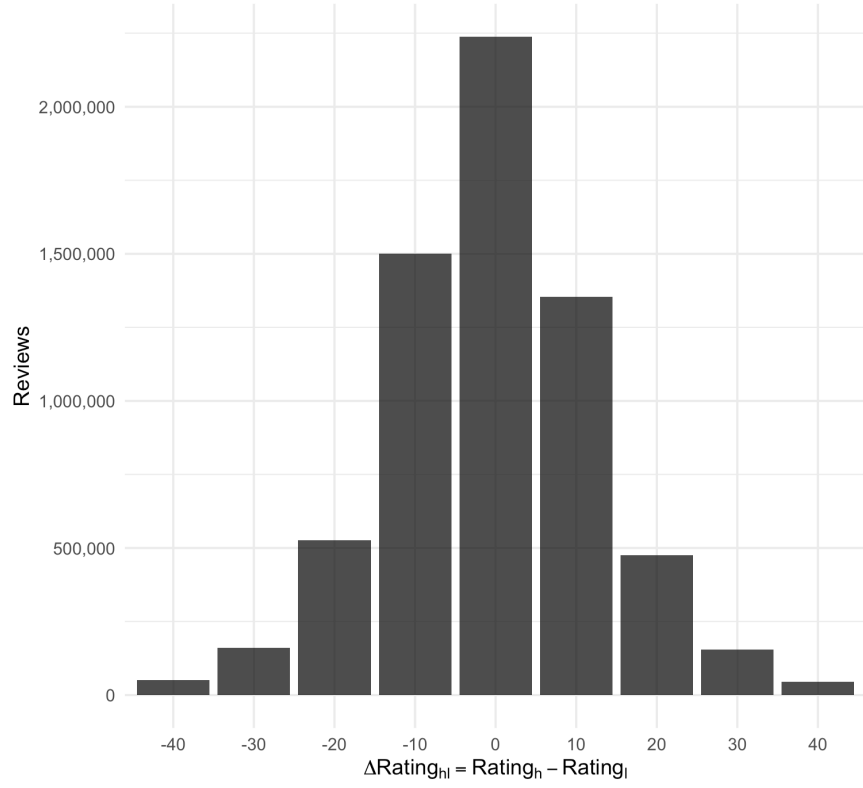


Figure 3: Distribution of rating differences across reviewer pairs

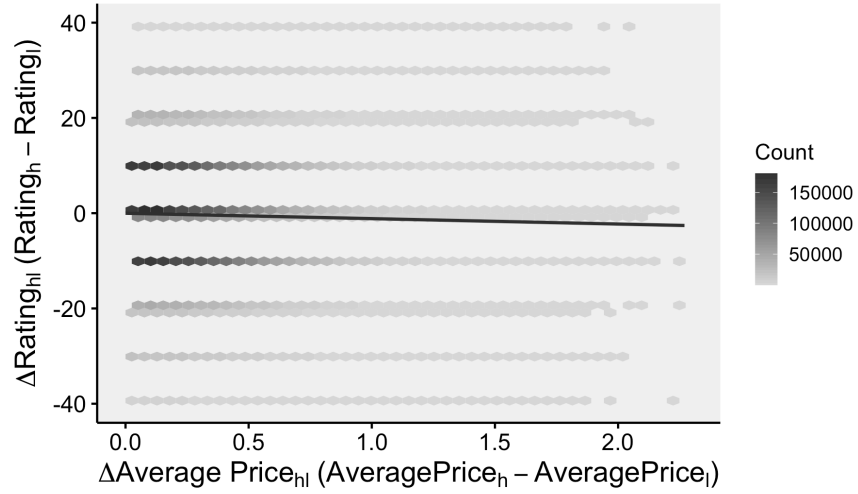


Figure 4: Relationship between all reviewer pairs through their average price differences and corresponding rating differences

$OnlineReputation_j$ refers to the restaurant's observed average rating on Tripadvisor at the time of a reviewer's visit. Thus, reputation captures the online evaluation signal available to consumers at the point when they reviewed the restaurant. Accounting for reputation changes is important because restaurants and their associated online reputations evolve over time. Prior research shows that online ratings significantly affect restaurant demand and revenues, suggesting that changes in observed reputation alter the information available to consumers (Luca, 2011). As a result, reviewers visiting the same restaurant at different points in time may evaluate not only different service experiences, but also different reputation signals visible on the platform. Because reviewers may observe different ratings of restaurant j and the time t they visit, differences in the restaurant's observed average rating at the time of each review are also controlled for:

$$\Delta OnlineReputation_j = OnlineReputation_{hj} - OnlineReputation_{lj}$$

where $OnlineReputation_{hj}$ denotes the restaurant's average rating at the time $User_h$ visits and $OnlineReputation_{lj}$ denotes the average rating at the time $User_l$ visits.

This variable is further decomposed into two components:

$$\Delta OnlineReputation_j^+ = \begin{cases} \Delta OnlineReputation_j & \text{if } \Delta OnlineReputation_j > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\Delta OnlineReputation_j^- = \begin{cases} \Delta OnlineReputation_j & \text{if } \Delta OnlineReputation_j < 0 \\ 0 & \text{otherwise} \end{cases}$$

Where, $\Delta OnlineReputation_j^+$ captures cases in which $User_h$ observes a higher restaurant reputation than $User_l$, while $\Delta OnlineReputation_j^-$ captures cases in which $User_h$ observes a lower reputation than $User_l$, independent of which reviewer visited the restaurant first. These variables control for differences in the online reputation signal faced by the two reviewers at the time of their visits.

To account for differences in visit timing, review dates are provided for each observation and

a binary variable $hFirst = 1$ if $User_h$ reviewed the restaurant at an earlier date than $User_l$, and 0 otherwise.

Each restaurant in the data has one of 3 possible price tiers. ‘\$’ represents the cheapest of establishments, ‘\$\$-\$\$\$’ represents the middle price tier, and ‘\$\$\$\$’ is assigned to the most expensive restaurants.

Table 1: Restaurant price tier statistics

Restaurant Price Category	Price Level	Restaurants	Reviewer Pairs
Cheap	\$	607	624,774
Mid-Range	\$\$-\$\$\$	1,978	4,758,852
Expensive	\$\$\$\$	143	1,119,967
Total		2,728	6,503,593

The sample is primarily composed of mid-range restaurants, which account for 1,978 establishments (72.5% of restaurants) and 4.75 million reviewer pairs (73.1% of all pairs). Budget restaurants represent 607 establishments (22.3%) and 624,774 reviewer pairs (9.6%), while premium restaurants account for only 143 establishments (5.2%) but generate over 1.1 million reviewer pairs (17.3%). This suggests that higher-priced restaurants receive substantially more reviews per establishment on average, generating a disproportionately large share of reviewer comparisons relative to their representation in the sample.

Table 2: Descriptive statistics

Variable	Mean	SD	Min	Max	N
$\Delta Rating_{hl}$	-0.447	13.101	-40.000	40.000	6,503,593
$\Delta AveragePrice_{hl}$	0.324	0.255	0.000	2.342	6,503,593
$\Delta OnlineReputation_j^+$	0.633	1.480	0.000	32.609	6,382,120
$\Delta OnlineReputation_j^-$	-0.634	1.480	-33.214	0.000	6,382,120
Days between reviews	605	461	0	2,554	6,503,593

Table 2 reports descriptive statistics for the main pair variables. On average, higher-average price reviewers assign slightly lower ratings than lower-average price reviewers evaluating the same restaurant. Reviewer pairs differ meaningfully in their average restaurant price levels, with an average difference of 0.447. The average temporal gap between reviews is approximately 605 days, further influencing the addition of reputation signal controls for this study.

RESULTS

Baseline Regression

We begin by examining whether differences in reviewers’ prior consumption experiences (average price levels) are associated with systematic differences in how the same restaurant is evaluated.

$$\Delta Rating_{hl,j} = \alpha + \beta \Delta AveragePrice_{hl} + \varepsilon_{hl,j} \quad (1)$$

Table 3: Difference in ratings due to difference in average price

Variable	Baseline (1)
Intercept	−0.037 (.03)
$\Delta AveragePrice_{hl}$	−1.265 (.16)***
R^2	0.00061
Adjusted R^2	0.00061
Observations (N)	6,503,593

Notes: Standard errors clustered at the restaurant level are reported in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table 3 reports the baseline relationship between differences in reviewers’ average prices and differences in the ratings they assign to the same restaurant. The coefficient on $\Delta AveragePrice_{hl}$ is negative and highly statistically significant ($\beta = -1.265, p < 0.001$), indicating that larger discrepancies in reviewers’ prior consumption experiences are associated with larger differences in restaurant evaluations. Specifically, a one-unit increase in the difference between reviewers’ average

restaurant price levels is associated with approximately a 1.26-point decrease in the higher-average price reviewer’s rating relative to the lower-average price reviewer evaluating the same restaurant.

While the coefficient may appear modest relative to the full 10–50 rating scale, it represents approximately 11.42% of the observed difference between restaurants at the 10th (rating of 33.69) and 90th (rating of 44.72) percentiles of average ratings. Given prior evidence that relatively small changes in online ratings can substantially affect restaurant demand (Anderson & Magruder, 2012), these differences are economically meaningful and are both likely to influence consumer decision-making as well as platform visibility.

These findings show that greater differences in reviewer consumption profiles lead to greater evaluative disagreement for the same restaurant. Higher-average price reviewers systematically assign lower ratings relative to lower-average price reviewers. The results support the argument that online ratings reflect heterogeneous evaluative standards and expectations rather than purely objective assessments of restaurant quality.

Including visit order and adding reputation signal changes

Next, we examine whether changes in visit timing and restaurant online rating changes help explain evaluative disagreement between reviewer types, since reviewer types may systematically differ in when they choose to visit restaurants according to the ratings and reviews posted online. To account for these dynamics, the following analysis incorporates changes in observed restaurant reputation as well as visit-order interactions to examine whether reviewer disagreement varies depending on how the observed restaurant reputation evolves over time.

Table 4: Reviewer visit order across restaurant reputation changes

Online Reputation Change	Reviewer h visits first	Reviewer h visits second
Online Reputation Increased	45.9%	54.1%
Online Reputation Decreased	55.3%	44.7%

There is a clear asymmetry in reviewer visit timing across restaurant reputation trajectories. When restaurant reputation improved over time, higher-average price reviewers were more likely

to visit after the increase (54.1%). In contrast, when restaurant reputation declined over time, higher-average price reviewers were substantially less likely to have visited after the decline occurred (44.7%).

$$hFirst_{ij} = \beta_0 + \beta_1 OnlineReputation_j^- + \varepsilon_{ij}$$

Table 5: Does visit order matter between reviewers?

Variables	Coefficient
	(1)
Intercept	0.498 (.003)***
$\Delta OnlineReputation^-$	-0.013 (.005)*
R^2	0.00152
Adjusted R^2	0.00152
Observations (N)	6,382,120

Notes: Standard errors clustered at the restaurant level are reported in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

The dependent variable equals one when the higher-average price reviewer visited the restaurant first and zero otherwise. The coefficient on $OnlineReputation_j^-$ ($\beta = -0.0103$) implies that $reviewer_h$ is more likely to have visited before a substantial deterioration in the restaurant's online reputation. Although the effect is statistically significant, its magnitude is economically small.

Higher-average price reviewers may be more likely to check and rely on online ratings prior to visiting and therefore become more deterred when a restaurant's reputation declines or falls below their expectations. In contrast, lower-average price reviewers may be less sensitive to declining reputation signals and therefore remain willing to visit restaurants even after online evaluations worsen. This interpretation is consistent with the idea that consumers with different prior consumption experiences may also differ in how strongly they respond to reputation information available on review platforms.

It is also important to incorporate controls for changes in restaurant reputation over time to examine whether evaluative disagreement reflects not only differences in reviewer standards, but also differences in the reputation signals observed by reviewers at the time of their visits. To account for these dynamics, the following regressions include separate controls for positive and negative changes in restaurant reputation between reviewer visits. This allows the analysis to examine whether improvements or declines in observed reputation systematically influence rating disagreement between reviewer types.

$$\begin{aligned}\Delta Rating_{hl,j} = & \alpha + \beta_1 \Delta AveragePrice_{hl} + \beta_2 \Delta OnlineReputation_j^+ \\ & + \beta_3 \Delta OnlineReputation_j^- + \varepsilon_{hl,j}\end{aligned}\tag{2}$$

$$\begin{aligned}\Delta Rating_{hl,j} = & \alpha + \beta_1 \Delta AveragePrice_{hl} + \beta_2 \Delta OnlineReputation_j^+ \\ & + \beta_3 \Delta OnlineReputation_j^- + \beta_4 hFirst_{hl,j} \\ & + \beta_5 (\Delta OnlineReputation_j^+ \times hFirst_{hl,j}) \\ & + \beta_6 (\Delta OnlineReputation_j^- \times hFirst_{hl,j}) + \varepsilon_{hl,j}\end{aligned}\tag{3}$$

Table 6 reports the regression results after accounting for changes in restaurant reputation and reviewer visit timing. Across all specifications, the coefficient on $\Delta AveragePrice_{hl}$ remains negative and highly statistically significant, ranging from -1.265 to -1.276. This stability suggests that the baseline relationship between reviewer consumption profiles and rating disagreement is robust to the inclusion of both reputation and temporal controls. Even after accounting for differences in the reputation signals observed by reviewers and the timing of their visits, higher-average price reviewers continue to assign systematically lower ratings than lower-average price reviewers evaluating the same restaurant. Column (2) introduces controls for positive and negative changes in restaurant reputation between reviewer visits. Both variables are negative and statistically significant, indicating that differences in observed reputation contribute to reviewer disagreement. However, the effect is notably stronger for reputation declines than for reputation increases.

Table 6: Accounting for observed reputation changes and visit order interactions

Variables	Baseline (1)	Observed Reputation Controls (2)	Visit Order Interactions (3)
Intercept	−0.037 (0.03)	−0.033 (0.03)	0.064 (0.07)
$\Delta AveragePrice$ ($AveragePrice_h - AveragePrice_l$)	−1.265 (0.16) ***	−1.276 (0.17) ***	−1.269 (0.17) ***
$OnlineReputation^+$ ($OnlineReputation_h > OnlineReputation_l$)		−0.152 (0.03) ***	−0.118 (0.04) **
$OnlineReputation^-$ ($OnlineReputation_h < OnlineReputation_l$)		−0.150 (0.03) ***	−0.253 (0.04) ***
$hFirst$ ($User_h$ reviewed first)			0.052 (0.11)
$OnlineReputation^+ \times hFirst$			−0.068 (0.06)
$OnlineReputation^- \times hFirst$			0.179 (0.06) **
Observations (N)	6,503,593	6,382,120	6,382,120
R^2	0.00061	0.00142	0.00069
Adjusted R^2	0.00061	0.00142	0.00069

Notes: Standard errors clustered at the restaurant level are reported in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Column (3) further incorporates visit-order interactions to examine whether the impact of reputation changes depends on which reviewer visited first. While the interaction between reputation increases and visit order is not statistically significant, the interaction between reputation decreases and visit order is positive and significant. This suggests that the effect of declining reputation on reviewer disagreement depends on when the higher-average price reviewer visited the restaurant.

The visit-order patterns reported earlier provide additional insight into this result. When restaurant reputation increased over time, higher-average price reviewers were more likely to visit

after the increase. In contrast, when reputation declined, higher-average price reviewers were substantially more likely to have visited before the decline occurred. Together, these findings suggest that higher-average price consumers may respond more strongly to online reputation signals when deciding whether to visit a restaurant. They appear less likely to visit restaurants following reputation deterioration and more likely to visit after positive reputation improvements. At the same time, the regression results indicate that when higher-average price reviewers do evaluate restaurants after a reputation decline, disagreement becomes particularly pronounced. One possible explanation is that these consumers are more sensitive to deteriorating reputation signals and therefore evaluate restaurants more critically when negative information is already visible on the platform.

Overall, these findings provide support that restaurant reputation dynamics and visit timing shape evaluative disagreement between reviewer types. More broadly, the results suggest that reviewer heterogeneity influences not only how consumers evaluate restaurants, but also when they choose to visit them. Consequently, disagreement in online ratings may reflect both differences in evaluative standards and differences in how consumers respond to evolving reputation signals.

Restaurant Customer Base Heterogeneity

The next step is to examine whether evaluative disagreement varies across restaurants serving different types of customer bases. While the previous analysis focuses on differences between reviewers themselves, disagreement may also depend on the broader composition of consumers attracted to a restaurant. Restaurants serving highly heterogeneous customer groups may attract consumers with substantially different prior experiences, potentially generating greater variation in evaluations. To investigate this possibility, the following analysis incorporates a restaurant-level measure of customer-base heterogeneity based on the dispersion of reviewers' average price levels within each restaurant.

For each restaurant, we calculate the standard deviation of all its reviewers average price levels ($SDAveragePrice_j$), where a high standard deviation means the restaurant attracts heterogeneous consumer segments, and a low standard deviation means it attracts a similar crowd all with similar average price levels. Across the 2,728 restaurants in the sample, the average customer

base heterogeneity is 0.2896, representing the standard deviation of the average price levels of each restaurant's reviewers. The standard deviation ($SDAveragePrice_j$) across restaurants is 0.0338, with a minimum value of 0.0462 and a maximum value of 0.5741.

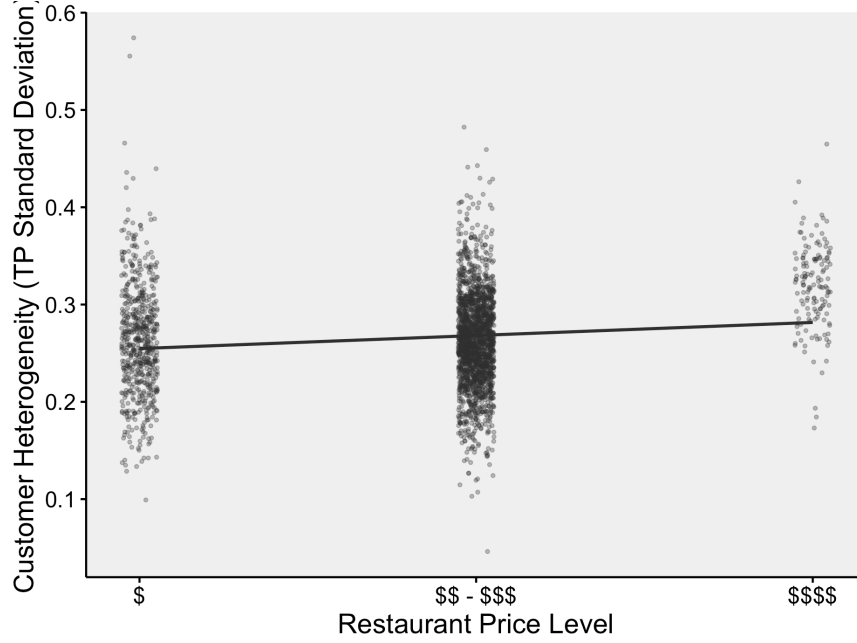


Figure 5: Customer heterogeneity in different restaurant price levels

Figure 5 illustrates how customer heterogeneity varies across restaurant price levels. The differences across categories are modest, but there. Substantial variation exists within each price level, where some restaurants attract highly heterogeneous consumers while others remain relatively homogeneous. For example, the most expensive restaurants attract a pretty diverse customer-base, whereas the cheapest restaurants have both establishments with low customer-base heterogeneity, and some with the highest of all the restaurants in the data.

This suggests that heterogeneity in consumer expectations is not driven solely by restaurant price, but rather by the types of customers a restaurant attracts. In particular, certain establishments appear to draw a broad mix of consumers with differing prior experiences.

$$\Delta Rating_{hl,j} = \alpha + \beta_1 \Delta AveragePrice_{hl} + \beta_2 SDAveragePrice_j + \varepsilon_{hl,j}$$

Table 7: Accounting for customer base heterogeneity by controlling for the standard deviation of all its reviewers’ average price levels

Variables	Customer-Base Heterogeneity
Intercept	−1.031 (0.34) **
$\Delta AveragePrice$ ($AveragePrice_h - AveragePrice_l$)	−1.329 (0.16) ***
Restaurant Customer-Base Heterogeneity ($SDAveragePrice_j$)	3.509 (1.17) **
Observations (N)	6,503,593
R^2	0.00069
Adjusted R^2	0.00069

Notes: Standard errors clustered at the restaurant level are reported in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table 7 examines whether customer-base heterogeneity contributes to reviewer disagreement. Consistent with previous findings, differences in reviewer average price profiles remain strongly associated with rating disagreement ($\beta = -1.329, p < 0.001$), meanwhile the coefficient on $SDAveragePrice_j$ is positive and statistically significant ($\beta = 3.509, p < 0.01$). This finding suggests that disagreement is driven not only by differences between reviewers themselves, but also by the broader composition of consumers a restaurant attracts. The results suggest that the negative relationship between reviewers’ average price levels and their ratings depends on the composition of the restaurant’s customer base. While reviewers who typically frequent higher-priced restaurants generally provide lower ratings than their lower-price counterparts, this difference narrows for restaurants that attract a more diverse mix of customers in terms of their usual spending levels. Restaurants that attract reviewers with a wide range of average price levels may offer more universally appealing experiences which may satisfy a broader range of expectations.

Differences Across Restaurant Categories

Table 8 below examines whether the relationship between reviewer average price differences and rating disagreement varies across restaurant price tiers.

$$\Delta Rating_{hl,j} = \alpha + \beta \Delta AveragePrice_{hl} + \varepsilon_{hl,j} \quad (1)$$

Table 8: The effect of average price differences on rating disagreements for different restaurant price tiers

Restaurant Tier	Intercept	$\Delta AveragePrice$ ($AveragePrice_h - AveragePrice_l$)	R^2
\$	−0.146 (0.08)	−1.820 (0.55) **	0.001 15
\$\$-\$\$\$	0.017 (0.04)	−1.530 (0.19) ***	0.000 84
\$\$\$\$	−0.103 (0.05) *	−0.295 (0.26)	0.000 04

Notes: Standard errors clustered at the restaurant level are reported in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

The negative association between $\Delta AveragePrice_{hl}$ and $\Delta Rating_{hl}$ is strongest among budget restaurants and remains significant for mid-range restaurants. For budget restaurants (\$), a one-unit increase in $\Delta AveragePrice_{hl}$ is associated with a 1.820-point decrease in $\Delta Rating_{hl}$. The effect is smaller for mid-range (\$\$-\$\$\$) restaurants, at 1.530 points. In contrast, no statistically significant relationship is observed for the most expensive (\$\$\$\$) restaurants. These findings suggest that higher-average price reviewers are substantially more critical than lower-average price reviewers when evaluating budget and, to a lesser extent, mid-range restaurants. However, reviewer differences in average spending behaviour do not translate into systematic differences in evaluations of most expensive restaurants, indicating that reviewer backgrounds matter less when assessing higher-end dining experiences. This ties back to the previous finding that restaurants which attract a wide clientele base have less significant rating disagreements, and the most expensive restaurants are those with the most customer heterogeneity as observed in Figure 5.

Table 9: Cuisine type categories

Primary Cuisine	Number of Restaurants	Number of Pairs	Average Pairs per Restaurant
Mediterranean	778	2,403,635	3,090
Seafood	140	671,573	4,797
Japanese	148	620,110	4,190
Bar	194	406,752	2,097
Spanish	210	366,501	1,745
Steakhouse	51	305,288	5,986
Italian	206	277,837	1,349
Cafés	112	121,146	1,082
European	74	109,974	1,486
Mexican	74	110,276	1,490
American	55	106,241	1,932
Other	686	1,004,260	1,464

Table 10 examines whether the relationship between reviewer average price differences and rating disagreement varies across primary cuisine types. The magnitude of the relationship differs across cuisines. The strongest negative association is observed for Italian restaurants, where a one-unit increase in $\Delta AveragePrice_{hl}$ is associated with a 2.48-point decrease in $\Delta Rating_{hl}$. Large and statistically significant negative effects are also observed for steakhouses, Spanish, seafood, and Mediterranean restaurants. In contrast, the relationship is statistically insignificant for Japanese restaurants, cafés, bars, Mexican restaurants, and American restaurants. This suggests that differences in reviewers’ average spending behaviour matter more when evaluating certain cuisines than others, indicating that the influence of prior dining experiences on restaurant evaluations is context dependent. One possible explanation is that different cuisine types attract systematically different customer populations. Some restaurant categories may appeal to a relatively homogeneous set of consumers with similar expectations and dining experiences, whereas others attract a broader mix of diners whose evaluative standards differ more substantially. As a result, the extent to which reviewers’ prior spending experiences influence their evaluations may vary across cuisine types.

$$\Delta Rating_{hl,j} = \alpha + \beta \Delta AveragePrice_{hl} + \varepsilon_{hl,j} \quad (1)$$

Table 10: The effect of reviewer average price differences on rating disagreement by cuisine type

Cuisine	Intercept	$\Delta AveragePrice$ ($AveragePrice_h - AveragePrice_l$)	R^2
Mediterranean	0.047 (0.02)	−1.231 (0.31) ***	0.000 60
Seafood	−0.084 (0.04)	−1.299 (0.40) **	0.000 63
Japanese	−0.013 (0.03)	−0.057 (0.60)	0.000 00
Bar	−0.152 (0.03)	−1.095 (0.67)	0.000 41
Spanish	0.014 (0.04)	−1.674 (0.61) **	0.001 21
Steakhouse	−0.222 (0.04)	−1.722 (0.35) ***	0.001 20
Italian	0.234 (0.04)	−2.481 (0.56) ***	0.001 87
Cafés	−0.174 (0.04)	−0.142 (0.47)	0.000 01
European	0.180 (0.05)	−1.547 (0.91)	0.000 86
Mexican	−0.188 (0.05)	−0.591 (0.98)	0.000 10
American	0.207 (0.06)	−0.276 (1.41)	0.000 02
Other	−0.090 (0.03)	−0.758 (0.35) *	0.000 20

Notes: Standard errors clustered at the restaurant level are reported in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

Overall, the findings indicate that reviewer heterogeneity does not affect all restaurant categories equally. The extent to which consumers disagree depends on the cuisine being evaluated, suggesting that some restaurant types generate greater variation in expectations and evaluative standards than others. This provides additional evidence that aggregate ratings may conceal important differences in how distinct customer segments perceive the same dining experience.

ROBUSTNESS CHECKS

To further investigate and support the effect of average price on rating differences, the following

robustness checks are performed:

1. Pairs with $\Delta AveragePrice_{hl} \geq 0.3$ only.
2. Reviewer demeaned ratings.
3. Reviewer pairs less than 365 days apart.
4. Analysis on the other 4 cities.

Pairs with $\Delta AveragePrice_{hl} \geq 0.3$ only

$$\Delta Rating_{hl,j} = \alpha + \beta \Delta AveragePrice_{hl} + \varepsilon_{hl,j} \quad (1)$$

$$\begin{aligned} \Delta Rating_{hl,j} = \alpha + \beta_1 \Delta AveragePrice_{hl} + \beta_2 \Delta OnlineReputation_j^+ \\ + \beta_3 \Delta OnlineReputation_j^- + \varepsilon_{hl,j} \end{aligned} \quad (2)$$

$$\begin{aligned} \Delta Rating_{hl,j} = \alpha + \beta_1 \Delta AveragePrice_{hl} + \beta_2 \Delta OnlineReputation_j^+ \\ + \beta_3 \Delta OnlineReputation_j^- + \beta_4 hFirst_{hl,j} \\ + \beta_5 (\Delta OnlineReputation_j^+ \times hFirst_{hl,j}) \\ + \beta_6 (\Delta OnlineReputation_j^- \times hFirst_{hl,j}) + \varepsilon_{hl,j} \end{aligned} \quad (3)$$

We restrict the sample to reviewer pairs with substantially different average price levels. This removes very similar pair comparisons according to their past consumption histories, and rather assess only those with significantly different average spending behaviours. Although this restriction removes nearly half of all reviewer pairs, the negative relationship between differences in reviewers' average price levels and evaluative disagreement remains statistically significant.

Table 11: Pairs with $\Delta AveragePrice_{hl} \geq 0.3$ only

Variables	Baseline (1)	Online Reputation Controls (2)	Visit Order Interactions (3)
Intercept	−0.080 (0.11)	−0.080 (0.12)	−0.031 (0.14)
$\Delta AveragePrice$ ($AveragePrice_h - AveragePrice_l$)	−1.031 (0.26) ***	−1.040 (0.27) ***	−1.036 (0.26) ***
$OnlineReputation^+$ ($OnlineReputation_h > OnlineReputation_l$)		−0.103 (0.04) **	−0.081 (0.05)
$OnlineReputation^-$ ($OnlineReputation_h < OnlineReputation_l$)		−0.109 (0.04) **	−0.260 (0.04) ***
$hFirst$ ($User_h$ reviewed first)			−0.114 (0.12)
$OnlineReputation^+ \times hFirst$			−0.042 (0.07)
$OnlineReputation^- \times hFirst$			0.251 (0.07) ***
R^2	0.00028	0.00059	0.00088
Adjusted R^2	0.00028	0.00059	0.00088
Observations (N)	3,429,000	3,372,418	3,372,418

Notes: Standard errors clustered at the restaurant level are reported in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Reviewer demeaned rating

To examine whether the results are driven by differences in reviewers' overall rating tendencies, ratings were demeaned by reviewer prior to constructing reviewer pairs. The negative relationship between reviewer average-price differences and evaluation differences remains highly significant after demeaning ($\beta = -1.970, p < 0.001$). Notably, the magnitude of the coefficient is larger than in the baseline specification using raw ratings ($\beta = -1.265$). This suggests that the main findings are not driven by higher-average price reviewers being systematically harsher reviewers overall. Instead, the results indicate that higher-average price reviewers evaluate the same restaurants less favorably

relative to their own rating standards, providing stronger evidence that reviewer consumption histories influence evaluations.

$$\Delta DemeanedRating_{hl,j} = \alpha + \beta \Delta AveragePrice_{hl} + \varepsilon_{hl,j}$$

Table 12: The relationship between reviewer average price differences and demeaned rating differences

Variables	Demeaned Ratings (1)
Intercept	0.014 (0.03)
$\Delta AveragePrice$ ($AveragePrice_h - AveragePrice_l$)	-1.970 (0.13) ***
R^2	0.00178
Adjusted R^2	0.00178
Observations (N)	6,503,593

Notes: Standard errors clustered at the restaurant level are reported in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. The dependent variable is the difference in reviewer-demeaned ratings between the higher-average price and lower-average price reviewer.

Reviewer pairs less than 365 days apart

To examine whether the main findings are driven by changes in restaurant quality over time, the analysis was repeated using only reviewer pairs whose visits occurred within 1 year of one another. The negative relationship between reviewer Average-price differences and rating disagreement remains highly significant ($\beta = -1.209, p < 0.001$) and is similar in magnitude to the baseline estimate ($\beta = -1.265$). The coefficient is remarkably stable despite eliminating over 62% of the reviewer pairs. This finding suggests that the results are not driven by long-term changes in restaurant quality or reputation, but instead reflect persistent differences in how higher-average price and lower-average price reviewers evaluate the same dining experiences.

$$\Delta Rating_{hl,j} = \alpha + \beta \Delta AveragePrice_{hl} + \varepsilon_{hl,j}, \quad \text{where } |date_h - date_l| \leq 365.$$

Table 13: Restricting reviewer pairs to reviews posted within 365 days

Variables	Within 365 Days (1)
Intercept	−0.052 (0.04)
$\Delta AveragePrice$ ($AveragePrice_h - AveragePrice_l$)	−1.209 (0.19) ***
R^2	0.00055
Adjusted R^2	0.00055
Observations (N)	2,447,646

Notes: Standard errors clustered at the restaurant level are reported in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Analysis including the four other cities

Table 14, below, provides summary statistics for the four other cities. Comparable findings between the cities are observed.

Table 14: Summary statistics including the other four cities (Madrid, Moscow, Istanbul, and Gijón)

Variable	Barcelona	Madrid	Moscow	Istanbul	Gijón
$\Delta Rating_{hl}$	−0.447 (13.13)	−0.531 (13.34)	−0.256 (13.29)	−0.118 (13.39)	−0.333 (13.26)
$\Delta AveragePrice_{hl}$	0.324 (0.26)	0.326 (0.26)	0.286 (0.22)	0.329 (0.26)	0.208 (0.16)
Customer-Base Heterogeneity ($SD AveragePrice_j$)	0.289 (0.03)	0.288 (0.04)	0.249 (0.04)	0.283 (0.05)	0.180 (0.03)
Reviewer $AveragePrice$	2.423 (0.33)	2.509 (0.33)	2.515 (0.29)	2.682 (0.34)	2.542 (0.19)
Restaurants (N)	2,728	3,693	2,083	938	265
Reviewers (N)	7,067	11,399	3,296	1,140	440
Reviews (N)	124,175	211,561	56,955	19,720	7,609
Reviewer pairs (N)	6,503,593	19,700,496	1,816,752	352,355	182,081

Notes: Means are reported first, with standard deviations in parentheses. Counts are reported for restaurants, reviewers, reviews, and reviewer pairs.

Before turning to the pairwise analysis, we first estimate a review-level regression using the original review data. This provides an alternative test of the relationship between reviewer *AveragePrice* and assigned ratings while controlling for restaurant and time fixed effects.

$$Rating_{ijt} = \beta_0 + \beta_1 AveragePrice_i + \alpha_j + \gamma_t + \varepsilon_{ijt}$$

Where i indexes reviewers, j indexes restaurants, and t indexes the year-month of the review. $Rating_{ijt}$ represents the rating assigned by reviewer i to restaurant j at time t , and $AveragePrice_i$ represents the average price level of restaurants previously reviewed by reviewer i . Restaurant fixed effects, α_j , control for time-invariant differences across restaurants, while year-month fixed effects, γ_t control for common temporal factors affecting ratings. The error term is represented by ε_{ijt} .

Table 15: Individual-level regression on all reviews

Variable	Barcelona	Madrid	Moscow	Istanbul	Gijón
<i>AveragePrice</i>	-1.326 (0.11) ***	-1.712 (0.10) ***	-1.338 (0.17) ***	-0.625 (0.25) *	-1.801 (0.79) *
R^2	0.121	0.131	0.077	0.093	0.137
Adjusted R^2	0.121	0.131	0.077	0.093	0.137
Observations (N)	124,175	211,561	56,955	19,720	7,609

Notes: The dependent variable is the restaurant rating assigned by each reviewer. Restaurant and time fixed effects are accounted for. Cluster-robust standard errors clustered at the restaurant level are reported in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

The results are broadly consistent across cities. Holding constant the restaurant being evaluated and common temporal factors, a one-unit increase in reviewer *AveragePrice* is associated with a statistically significant decrease in the assigned restaurant rating in Barcelona, Madrid, Moscow, Istanbul, and Gijón. The estimated effects range from a 0.62-point decrease in Istanbul to a 1.80-point decrease in Gijon.

Overall, these findings are consistent with the main pairwise analysis, in which larger differences in *AveragePrice* between reviewers were associated with lower relative ratings assigned by higher *AveragePrice* reviewers. The consistency in the direction and magnitude of the estimated effects across both the review-level and pairwise analyses provides additional evidence that reviewers with higher *AveragePrice* profiles tend to evaluate the same restaurants less favourably than reviewers with lower *AveragePrice* profiles.

Next, we run the same regression analyses as on the 4 other cities where we observe familiar results.

$$\Delta Rating_{hl,j} = \alpha + \beta \Delta AveragePrice_{hl} + \varepsilon_{hl,j} \quad (1)$$

$$\begin{aligned} \Delta Rating_{hl,j} = \alpha + \beta_1 \Delta AveragePrice_{hl} + \beta_2 \Delta OnlineReputation_j^+ \\ + \beta_3 \Delta OnlineReputation_j^- + \varepsilon_{hl,j} \end{aligned} \quad (2)$$

$$\begin{aligned} \Delta Rating_{hl,j} = \alpha + \beta_1 \Delta AveragePrice_{hl} + \beta_2 \Delta OnlineReputation_j^+ \\ + \beta_3 \Delta OnlineReputation_j^- + \beta_4 hFirst_{hl,j} \\ + \beta_5 (\Delta OnlineReputation_j^+ \times hFirst_{hl,j}) \\ + \beta_6 (\Delta OnlineReputation_j^- \times hFirst_{hl,j}) + \varepsilon_{hl,j} \end{aligned} \quad (3)$$

$$\begin{aligned} \Delta Rating_{hl,j} = \alpha + \beta_1 \Delta AveragePrice_{hl} + \beta_2 SDAveragePrice_j \\ + \varepsilon_{hl,j} \end{aligned} \quad (4)$$

$$\Delta DemeanedRating_{hl,j} = \alpha + \beta \Delta AveragePrice_{hl} + \varepsilon_{hl,j} \quad (5)$$

$$\Delta Rating_{hl,j} = \alpha + \beta \Delta AveragePrice_{hl} + \varepsilon_{hl,j}, \quad (6)$$

where $|date_h - date_l| \leq 365$.

For **Tables 16-19** below:

$\Delta AveragePrice$ denotes the difference between the average price levels of the higher- and lower-priced reviewers in each pair ($AveragePrice_h - AveragePrice_l$). $Reputation^+$ measures the extent to which the restaurant's online reputation, based on all reviews posted prior to user h 's visit, exceeded the reputation observed by user h . Conversely, $Reputation^-$ measures the extent to which the restaurant's online reputation, based on all reviews posted prior to user l 's visit, exceeded the reputation observed by user l . $hFirst$ is an indicator equal to 1 if user h visited the restaurant before user l . Customer-Base Heterogeneity ($SDAveragePrice_j$) is the standard deviation of the average price levels of all reviewers who reviewed restaurant j . Higher values indicate that the restaurant attracts reviewers with more heterogeneous average consumption profiles, whereas lower values indicate a more homogeneous customer base.

Table 16: Analysis on Madrid

Variables	Baseline (1)	Reputation Controls (2)	Visit Order (3)	Customer-Base Heterogeneity (4)	Demeaned Ratings (5)	Within 365 Days (6)
Intercept	-0.114 (0.02) ***	-0.129 (0.03) ***	-0.142 (0.12)	-0.997 (0.26) ***	-0.044 (0.02) *	-0.088 (0.03) **
$\Delta AveragePrice$ ($AveragePrice_h - AveragePrice_l$)	-1.280 (0.19) ***	-1.275 (0.20) ***	-1.272 (0.19) ***	-1.363 (0.18) ***	-2.448 (0.24) ***	-1.433 (0.18) ***
$Reputation^+$		-0.055 (0.04)	-0.038 (0.03)			
$Reputation^-$		-0.082 (0.03) **	-0.149 (0.05) ***			
$hFirst$			0.024 (0.24)			
$Reputation^+ \times hFirst$			-0.035 (0.07)			
$Reputation^- \times hFirst$			0.112 (0.06)			
Customer-Base Heterogeneity ($SDAveragePrice_j$)				3.161 (0.91) ***		
Observations (N)	19,700,496	19,496,212	19,496,212	19,700,496	19,700,496	7,998,531
R^2	0.00061	0.00070	0.00073	0.00070	0.00270	0.00076
Adjusted R^2	0.00061	0.00070	0.00073	0.00070	0.00270	0.00076

Notes: Cluster-robust standard errors clustered at the restaurant level are reported in parentheses. ***

$p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

Table 17: Analysis on Moscow

Variables	Baseline (1)	Reputation Controls (2)	Visit Order (3)	Customer-Base Heterogeneity (4)	Demeaned Ratings (5)	Within 365 Days (6)
Intercept	0.147 (0.05) **	-0.924 (0.83)	6.897 (2.18) **	0.356 (0.40)	-0.006 (0.05)	0.074 (0.06)
$\Delta AveragePrice$ ($AveragePrice_h - AveragePrice_l$)	-1.409 (0.21) ***	-1.376 (0.21) ***	-1.313 (0.22) ***	-1.385 (0.20) ***	-2.241 (0.21) ***	-1.237 (0.25) ***
$Reputation^+$		-0.158 (0.04) ***	-0.364 (0.07) ***			
$Reputation^-$		0.183 (0.04) ***	0.186 (0.03) ***			
$hFirst$			-17.019 (3.59) ***			
$Reputation^+ \times hFirst$			0.145 (0.06) *			
$Reputation^- \times hFirst$			0.294 (0.05) ***			
Customer-Base Heterogeneity ($SDAveragePrice_j$)				-0.864 (1.62)		
Observations (N)	1,816,752	1,763,744	1,763,744	1,816,752	1,816,752	756,731
R^2	0.00055	0.00159	0.00736	0.00056	0.00174	0.00043
Adjusted R^2	0.00055	0.00159	0.00736	0.00056	0.00174	0.00042

Notes: Cluster-robust standard errors clustered at the restaurant level are reported in parentheses. ***

$p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

Table 18: Analysis on Istanbul

Variables	Baseline (1)	Reputation Controls (2)	Visit Order (3)	Customer-Base Heterogeneity (4)	Demeaned Ratings (5)	Within 365 Days (6)
Intercept	0.135 (0.09)	-1.225 (0.88)	13.480 (1.81) ***	0.193 (0.53)	0.037 (0.08)	0.168 (0.09) .
$\Delta AveragePrice$ ($AveragePrice_h - AveragePrice_l$)	-0.768 (0.29) **	-0.689 (0.29) *	-0.684 (0.30) *	-0.759 (0.27) **	-0.577 (0.28) *	-0.894 (0.33) **
$Reputation^+$		-0.274 (0.04) ***	-0.613 (0.05) ***			
$Reputation^-$		0.308 (0.05) ***	0.275 (0.05) ***			
$hFirst$			-30.201 (2.92) ***			
$Reputation^+ \times hFirst$			0.293 (0.05) ***			
$Reputation^- \times hFirst$			0.468 (0.06) ***			
Customer-Base Heterogeneity ($SDAveragePrice_j$)				-0.214 (1.86)		
Observations (N)	352,355	335,197	335,197	352,355	352,355	119,908
R^2	0.00021	0.00481	0.01406	0.00021	0.00015	0.00028
Adjusted R^2	0.00021	0.00480	0.01404	0.00021	0.00015	0.00028

Notes: Cluster-robust standard errors clustered at the restaurant level are reported in parentheses. ***

$p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

Table 19: Analysis on Gijón

Variables	Baseline (1)	Reputation Controls (2)	Visit Order (3)	Customer-Base Heterogeneity (4)	Demeaned Ratings Within 365 Days (5)	(6)
Intercept	0.494 (0.20) *	-0.563 (1.63)	3.980 (2.83)	0.829 (1.15)	-0.006 (0.18)	0.558 (0.25) *
$\Delta AveragePrice$ ($AveragePrice_h - AveragePrice_l$)	-3.974 (1.72) *	-4.082 (1.76) *	-4.039 (1.76) *	-3.933 (1.65) *	-4.053 (1.51) **	-4.391 (2.03) *
$Reputation^+$		-0.318 (0.06) ***	-0.428 (0.09) ***			
$Reputation^-$		0.345 (0.07) ***	0.344 (0.06) ***			
$hFirst$			-8.494 (3.72) *			
$Reputation^+ \times hFirst$			0.122 (0.10)			
$Reputation^- \times hFirst$			0.086 (0.09)			
Customer-Base Heterogeneity ($SDAveragePrice_j$)				-1.906 (5.74)		
Observations (N)	182,081	175,130	175,130	182,081	182,081	68,953
R^2	0.00238	0.00762	0.00877	0.00240	0.00304	0.00290
Adjusted R^2	0.00238	0.00760	0.00873	0.00238	0.00303	0.00288

Notes: Cluster-robust standard errors clustered at the restaurant level are reported in parentheses. ***

$p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

DISCUSSION

This study examines whether consumers with different consumption histories evaluate the same restaurant differently. Using a reviewer-pair design that compares higher-average price and lower-average price reviewers evaluating the same restaurant, the results reveal a consistent pattern of disagreement.

Across reviewer pairs, higher-average price reviewers systematically assigned lower ratings than lower-average price reviewers, and the magnitude of disagreement increased as differences in reviewer average price widened. The strength of this relationship also varied across restaurant characteristics, with the largest effects observed for the cheapest price-tier restaurants and for Italian cuisine.

Restaurants attracting consumers with widely varying consumption profiles exhibited smaller differences in evaluations than restaurants serving more homogeneous customer groups for Barcelona and Madrid which are the two cities with the most reviews, but remain insignificant for the other cities, potentially reflecting their substantially smaller sample sizes. The main

finding remained robust after controlling for online restaurant reputation signal changes, using reviewer-demeaned ratings, restricting analyses to reviews posted within one year of one another, and on four other cities.

One interpretation of the findings is that differences in consumption histories create differences in evaluative standards. Consumers accustomed to higher-priced restaurants may place greater emphasis on service quality, atmosphere, presentation, authenticity, aesthetics, or other attributes that are less salient to consumers with lower-priced consumption histories (Adhikari, 2015; Cho et al., 2024). Consequently, when two consumers evaluate the same restaurant, they may focus on different dimensions of the experience and arrive at different conclusions regarding quality.

Theoretical Implications

The results highlight that consumer evaluations are shaped not only by product quality, but also by the composition of the customer base. These results provide support for expectation-disconfirmation theory and reference point theory in the context of online reviews. Consumers do not appear to evaluate restaurants using a common standard. Instead, evaluations are shaped by prior consumption experiences, which influence the expectations and reference points consumers bring to a dining experience. Higher-average price reviewers apply more demanding standards when evaluating the same establishments as lower-average price reviewers. The study also contributes to the online review literature by demonstrating that disagreement is systematic rather than random, and through the findings it is suggested that aggregate ratings conceal meaningful variation in how different consumer segments evaluate the same experience. Consequently, disagreement itself may contain valuable information about consumer heterogeneity that is lost when reviews are aggregated into a single average score. More broadly, the results challenge the implicit assumption that online ratings represent objective measures of quality. Instead, ratings appear to reflect an interaction between restaurant characteristics and reviewer-specific evaluative standards. The same restaurant may therefore be perceived quite differently depending on the experiences and expectations of the consumer providing the evaluation.

Implications for Consumers

Research shows that over 90% of consumers use the average rating to choose a restaurant (Dai, Jin, Lee, & Luca, 2012). In doing so, they often assume that other consumers apply evaluative standards similar to their own via a false consensus bias (Pope, 2013; Bauman & Geher, 2002). As a result, ratings are frequently interpreted as objective indicators of quality rather than as reflections of individual experiences and expectations. The findings of this study suggest that such an assumption may be problematic. Reviews and ratings are generated by individuals with diverse consumption histories and reference points. Consequently, a rating may reflect the standards of the reviewer providing the evaluation. When consumers rely on evaluations written by reviewers whose experiences differ substantially from their own, they may encounter outcomes that deviate from their expectations. This can result in experiences being perceived as either better or worse than anticipated, leading to subsequent ratings that further reinforce variation in aggregate evaluations.

Implications for Managers and Firms

For managers, the findings suggest a trade-off between attracting a broad customer base and maintaining consistent evaluations. While appealing to a diverse range of consumers may increase demand, it may also introduce greater variation in expectations and evaluative standards. Consistent with the results for customer-base heterogeneity, restaurants attracting consumers with more diverse consumption profiles exhibited greater disagreement in ratings. This finding implies that businesses serving highly heterogeneous customer segments may face greater rating volatility, even when objective service quality remains unchanged. Differences in customer expectations can produce divergent evaluations of the same experience, leading to less stable online reputations. In contrast, businesses with more narrowly defined target markets may experience greater consistency in customer evaluations because consumers arrive with more similar expectations and reference points. Managers may benefit from looking beyond average ratings and examining the composition of their reviewers. Understanding which types of consumers are attracted to a business can help managers better align pricing, branding, positioning, and marketing communications with customer expectations. By reducing expectation differences, firms may be able to improve customer satisfaction and generate more consistent evaluations over time.

Implications for Online Review Platforms

The results also have implications for the design of online review platforms. Most platforms currently aggregate evaluations from all consumers into a single rating, implicitly treating reviews as equally informative for all users. As these ratings influence product visibility, rankings, and consumer choice on digital platforms (Motoyama and Usher, 2020; Fang, 2022), disagreement among reviewers can ultimately affect the success and visibility of products and services. One potential response is the development of more personalized review and recommendation systems. Rather than relying exclusively on aggregate ratings, platforms could place greater emphasis on evaluations from reviewers who share similar characteristics, preferences, or consumption histories with the focal user. Some online platforms such as Expedia have already introduced an “others like you” recommendation system which helps similar consumers see other people like them (Lewis and Zervas, 2019), but despite these social features, the aggregate rating is consistently identified as the most salient and influential piece of information for the majority of users (Dai et al., 2012). A further approach on this idea may provide consumers with information that is more relevant to their own expectations and improve the predictive value of online reviews.

More broadly, the findings suggest that reviewer heterogeneity represents an important source of information rather than simply noise in the rating process. Investments in identifying which characteristics different reviewers prefer and evaluate may therefore improve both recommendation quality and consumer satisfaction.

LIMITATIONS AND FUTURE RESEARCH

This study has several limitations that provide opportunities for future research.

First, reviewer consumption histories are measured using the average price level of restaurants previously visited. While this measure captures meaningful differences in prior consumption experiences, it represents only one dimension of consumer heterogeneity. Consumers may also differ in terms of the cuisines they typically visit, their overall reviewing behaviour, their level of experience on the platform, or other characteristics that are not captured by price alone. Because demographic information is not available in the data, the analysis cannot examine how factors such as age, income, education, or cultural background contribute to evaluative disagreement. Future research could explore alternative measures of reviewer experience and investigate whether

other dimensions of consumption history generate similar patterns of disagreement.

Second, while the study identifies systematic disagreement between higher-average price and lower-average price reviewers, it cannot directly observe the mechanisms underlying these differences. The results are consistent with differences in expectations and reference points, but the data does not reveal which aspects of the dining experience drive disagreement. Future research should analyze review text alongside the given rating to examine whether different types of reviewers focus on distinct attributes when forming evaluations. It would be interesting to see how much emphasis is put on different attributes such as presentation, authenticity, value, portion size, or convenience. Understanding these underlying mechanisms would provide a richer causal explanation for why disagreement emerges.

Third, the analysis focuses on a single platform and looks at five European cities. Although the reviewer-pair methodology could be applied more broadly, the extent to which these findings generalize to other cultural contexts, industries, and review platforms remains an open question. Future research could examine whether similar patterns emerge across different countries, digital review environments, and of course product categories.

Overall, the findings suggest that online ratings reflect more than objective assessments of quality. Rather, evaluations are shaped by consumers' prior experiences and the standards they use when interpreting those experiences. As online reviews continue to influence consumer decision-making and firm performance, understanding the sources of reviewer disagreement becomes increasingly important for researchers, managers, digital platforms, and consumers alike.

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